

ECON 695 – Problem Set 5

Due: November 3, 2025

AI Disclaimer

I used AI tools (ChatGPT) to help troubleshoot minor LaTeX and Python issues, to reorganize sections for clarity (namely indentation conventions and organization in the `PS5_twins.ipynb` file), and to draft descriptive tables (because they are a pain in LaTeX). All analysis, model specification, interpretation, and final write-up are my own.

Twins Study (Princeton Twins Survey)

In this problem set I work with the Princeton Twins Survey (`twins.csv`). Each observation is a twin from a pair of identical twins, indexed by a family id i and a twin indicator $t \in \{1, 2\}$. Below I restate the variable dictionary from the assignment for ease of reference:

- `famid` = family id variable.
- `t` = 1, 2 for twin #1 or twin #2.
- `age` = age (some observations include fractional years).
- `educ` = education (years).
- `oeduc` = education of the other twin.
- `lw` = log wage.
- `married` = 1 if married, 0 otherwise.
- `omarried` = marriage status of the other twin.
- `female` = 1 if female.
- `ofemale` = 1 if the other twin is female.

- `exp` = “labor market experience” = `age` − `educ` − 6.
- `oexp` = experience of the other twin.

Note. All pairs in the file are identical twins, so `female` = `ofemale` for every observation.

Data setup & quick checks

Before starting Question 1, I performed basic checks in the notebook:

- 1) Uniqueness: (`famid`, `t`) combinations are unique (one row per twin).
- 2) Sex consistency: `female` = `ofemale` holds for all pairs (same sex by design).
- 3) Feature engineering: I created `exp2` = `exp`² for use in quadratic experience profiles.

Descriptive statistics (key variables)

I summarize several variables that are used repeatedly below (sample size $n = 1,074$). The values match the notebook output produced via `pandas.describe()`.

Table 1: Summary statistics

Variable	count	mean	std	min	25%	50%	75%
max							
<code>lw</code>	1074	2.4710	0.6098	0.7226	2.0441	2.4571	2.8212
4.8929							
<code>educ</code>	1074	14.0805	2.0298	8.0000	12.0000	14.0000	16.0000
18.0000							
<code>exp</code>	1074	17.8336	11.8061	−2.0000	9.0000	16.7361	25.0000
53.0000							
<code>exp2</code>	1074	457.2928	518.2904	0.0000	81.0000	280.1114	625.0000
2809.0000							
<code>married</code>	1074	0.4795	0.4998	0.0000	0.0000	0.0000	1.0000
1.0000							
<code>female</code>	1074	0.5345	0.4990	0.0000	0.0000	1.0000	1.0000
1.0000							

Q1. Simple Model: Log Wages on Schooling, Experience, Marital Status, and Sex

Estimate a simple model relating log wages to: education, experience, experience-squared, married status, and female.

Model

I estimate the OLS model

$$\underbrace{\ln w_{it}}_{\log(\text{wage})} = \beta_0 + \beta_1 \text{educ}_{it} + \beta_2 \text{exp}_{it} + \beta_3 \text{exp}_{it}^2 + \beta_4 \text{married}_{it} + \beta_5 \text{female}_{it} + \varepsilon_{it},$$

using heteroskedasticity-robust (HC1) standard errors. The marginal effect of experience is

$$\frac{\partial \ln w_{it}}{\partial \text{exp}_{it}} = \beta_2 + 2\beta_3 \text{exp}_{it}.$$

Results

Table 2: Baseline OLS with HC1 standard errors				
Variable	Coefficient	Robust SE	t	p
Intercept	0.3455	0.1386	2.49	0.013
Education	0.1238	0.0086	14.36	0.000
Experience	0.0523	0.0048	10.88	0.000
Experience ²	-0.0009	0.0001	-8.28	0.000
Married	0.0693	0.0321	2.16	0.031
Female	-0.3341	0.0309	-10.83	0.000
$N = 1,074; \quad R^2 = 0.3146; \quad \bar{R}^2 = 0.3113.$				

Interpretation

- Schooling: an additional year of education raises expected log wages by about 0.124 log points; using the exact transformation, this is $100 \times (\exp(0.1238) - 1) \approx 13.2\%$.
- Experience: wages rise with experience but at a diminishing rate because $\beta_3 < 0$. The implied turning point is

$$\text{exp}^* = \frac{\beta_2}{-2\beta_3} = \frac{0.0523}{2(0.0009)} \approx 29.1 \text{ years},$$

beyond which additional experience adds little to wages on average.

- Marital status: being married is associated with roughly $100 \times (\exp(0.0693) - 1) \approx 7.2\%$ higher wages, significant at the 5% level.
- Sex: the coefficient on `female` is -0.334 ; in levels this corresponds to $100 \times (\exp(-0.3341) - 1) \approx -28.4\%$, i.e., women earn about 28% less than men on average, conditional on co-variates.
- Model fit: the model explains about 31% of the variation in $\log(\text{wage})$ ($R^2 = 0.3146$), which is reasonable for a cross-sectional wage regression.

Q2. Separate Models for Men and Women

Fit the same model separately for men and women. Does the model look different?

Specification

For each gender $g \in \{\text{men, women}\}$ I estimate

$$\ln w_i = \beta_{0g} + \beta_{1g} \text{educ}_i + \beta_{2g} \text{exp}_i + \beta_{3g} \text{exp}_i^2 + \beta_{4g} \text{married}_i + u_i,$$

using HC1 robust standard errors.

Results

Table 3: Men: OLS with HC1 robust standard errors

Variable	Coef.	SE(HC1)	<i>t</i>	<i>p</i>
Intercept	0.5197	0.2194	2.3685	0.0179
Education	0.1027	0.0135	7.5896	0.0000
Experience	0.0588	0.0080	7.3951	0.0000
Experience ²	-0.0010	0.0002	-5.8950	0.0000
Married	0.1993	0.0506	3.9419	0.0001
$N = 500$; $R^2 = 0.2553$; $\bar{R}^2 = 0.2493$.				

Table 4: Women: OLS with HC1 robust standard errors

Variable	Coef.	SE(HC1)	t	p
Intercept	-0.0854	0.1595	-0.5353	0.5924
Education	0.1369	0.0101	13.5113	0.0000
Experience	0.0529	0.0059	8.9736	0.0000
Experience ²	-0.0010	0.0001	-6.8237	0.0000
Married	-0.0399	0.0407	-0.9813	0.3264
$N = 574$; $R^2 = 0.3033$; $\bar{R}^2 = 0.2984$.				

Turning points

With the quadratic experience term, the implied turning point is

$$\text{exp}_g^* = \frac{\beta_{2g}}{-2\beta_{3g}}.$$

Plugging in the estimates gives

$$\text{Men: } \text{exp}^* \approx 29.8 \text{ years} \quad \text{and} \quad \text{Women: } \text{exp}^* \approx 26.9 \text{ years}.$$

Does the model look different?

- **Education:** Positive and highly significant for both; the effect is *larger for women* (0.137 vs. 0.103).
- **Experience profile:** Positive β_2 and negative β_3 in both groups (concave), with a slightly earlier peak for women (26.9 vs. 29.8 years).
- **Married:** *Men:* positive and significant (≈ 0.20). *Women:* small, negative, and insignificant (-0.040).
- **Fit:** R^2 is a bit higher for women (0.303) than men (0.255).

Q3. Mean Family Marriage Rate

Construct the mean family marriage rate for each person in the data set (i.e., the fraction of the twins that is married, which can be 0, 1/2, or 1).

Part (a): Verifying the Auxiliary Regression

We define the mean family marriage rate for each family f as the average of the twins' marriage indicators:

$$\text{mean_married}_f = \frac{1}{2} \sum_{t=1}^2 \text{married}_{ft},$$

which can take values 0, 0.5, or 1, depending on whether neither, one, or both twins are married.

We then estimate the auxiliary regression

$$\text{married}_{ft} = \alpha + \rho \text{mean_married}_f + \varepsilon_{ft},$$

using HC1 robust standard errors.

Table 5: Auxiliary regression: verifying the mean-marriage relationship

Variable	Coefficient	Robust SE (HC1)	t	p
Intercept	0.0000	0.0074	0.0000	1.0000
Mean Married	1.0000	0.0009	1147.51	0.0000
$N = 1,074; \quad R^2 = 0.7370; \quad \bar{R}^2 = 0.7367.$				

Interpretation

The estimated coefficient on `mean_married` is essentially 1, confirming the theoretical result discussed in Lecture 8. Intuitively, each twin's marital status is mechanically tied to the family's mean marriage rate: if one twin's status changes (from 0 to 1, or vice versa), the family average changes in the same direction by exactly half for both siblings, implying a perfect one-to-one mapping in the regression.

Mathematically, for any twin t in family f ,

$$\text{mean_married}_f = \frac{1}{2} (\text{married}_{f1} + \text{married}_{f2}) = \frac{1}{2} \text{married}_{ft} + \frac{1}{2} \text{married}_{f(-t)}.$$

Because `mean_marriedf` contains `marriedft` as part of its own definition, regressing `marriedft` on `mean_marriedf` yields an estimated slope of 1 by construction: any within-family change in `marriedft` moves both sides of the equation equally.

Empirically, this is reflected in the near-perfect fit ($R^2 \approx 0.74$) and a coefficient statistically indistinguishable from 1.

Part (b): Adding the family mean marriage rate to gender-specific wage models

We augment the Q2 models to include the family mean marriage rate `mean_marriedf` and estimate, separately by gender,

$$\ln w_i = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exp}_i + \beta_3 \text{exp}_i^2 + \beta_4 \text{married}_i + \beta_5 \text{mean_married}_f + u_i,$$

using HC1 robust standard errors.

Table 6: Men: OLS with `mean_married` (HC1 robust)

Variable	Coef.	SE(HC1)	<i>t</i>	<i>p</i>
Intercept	0.5102	0.2200	2.3187	0.0204
Education	0.1029	0.0135	7.6062	0.0000
Experience	0.0572	0.0079	7.1902	0.0000
Experience ²	−0.0010	0.0002	−5.8788	0.0000
Married	0.0300	0.0897	0.3339	0.7385
Mean married	0.2500	0.1108	2.2571	0.0240
<i>N</i> = 500; <i>R</i> ² = 0.2619; $\bar{R}^2$ = 0.2544.				

Table 7: Women: OLS with `mean_married` (HC1 robust)

Variable	Coef.	SE(HC1)	<i>t</i>	<i>p</i>
Intercept	−0.0826	0.1597	−0.5173	0.6050
Education	0.1368	0.0102	13.4580	0.0000
Experience	0.0534	0.0062	8.6071	0.0000
Experience ²	−0.0010	0.0001	−6.6896	0.0000
Married	−0.0221	0.0728	−0.3031	0.7618
Mean married	−0.0268	0.0915	−0.2930	0.7695
<i>N</i> = 574; <i>R</i> ² = 0.3034; $\bar{R}^2$ = 0.2972.				

Table 8: Marriage coefficients before (Q2) vs after adding `mean_married` (Q3b)

Group	Married (Q2)	Married (Q3b)	Mean married	Δ Married (Q3b−Q2)
Men	0.1993	0.0300	0.2500	−0.1693
Women	−0.0399	−0.0221	−0.0268	0.0179

What changed?

- **Men.** The individual marriage coefficient shrinks from ≈ 0.199 (Q2) to ≈ 0.030 (Q3b) and becomes insignificant, while the family mean term is positive and significant ($\hat{\beta}_5 \approx 0.25$).
- **Women.** The marriage coefficient was already small and insignificant in Q2 (≈ -0.040) and remains small/insignificant in Q3b; the family mean term is also small and insignificant.

Interpretation (men vs. women)

With two siblings per family, $\text{mean_married}_f = (\text{married}_{f1} + \text{married}_{f2})/2$ captures a *family-level (shared) component* of marriage—selection or common factors that influence both twins (values, timing, labor-market conditions). For men, most of the apparent “marriage premium” in Q2 is absorbed by this shared component (strong $\hat{\beta}_5 > 0$), leaving little individual-specific premium after controlling for the family mean. For women, there is no evident premium to begin with, and controlling for the family mean changes little—both the individual and family components are economically small and statistically insignificant.

Model fit

Fit is essentially unchanged for women ($R^2 \approx 0.303 \rightarrow 0.303$) and rises only slightly for men ($0.255 \rightarrow 0.262$), consistent with explanatory power from the family mean operating mainly in the male sample.

Part (c): Using the co-twin’s marriage status instead of the family mean

We re-estimate the gender-specific models from Q2 replacing the family mean term with the other twin’s marriage status omarried_i :

$$1w_i = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exp}_i + \beta_3 \text{exp}_i^2 + \beta_4 \text{married}_i + \beta_5 \text{omarried}_i + u_i,$$

using HC1 robust SEs.

Comparison and interpretation

Link between (b) and (c). With two siblings per family,

$$\text{mean_married}_f = \frac{1}{2}(\text{married}_{f1} + \text{married}_{f2}) = \frac{1}{2}\text{married}_i + \frac{1}{2}\text{omarried}_i.$$

Table 9: Men: OLS with `omarried` (HC1 robust)

Variable	Coef.	SE(HC1)	t	p
Intercept	0.5102	0.2200	2.3187	0.0204
Education	0.1029	0.0135	7.6062	0.0000
Experience	0.0572	0.0079	7.1902	0.0000
Experience ²	-0.0010	0.0002	-5.8788	0.0000
Married	0.1550	0.0539	2.8771	0.0040
Other twin married	0.1250	0.0554	2.2571	0.0240

$N = 500$; $R^2 = 0.2619$; $\bar{R}^2 = 0.2544$.

Table 10: Women: OLS with `omarried` (HC1 robust)

Variable	Coef.	SE(HC1)	t	p
Intercept	-0.0826	0.1597	-0.5173	0.6050
Education	0.1368	0.0102	13.4580	0.0000
Experience	0.0534	0.0062	8.6071	0.0000
Experience ²	-0.0010	0.0001	-6.6896	0.0000
Married	-0.0355	0.0432	-0.8202	0.4121
Other twin married	-0.0134	0.0457	-0.2930	0.7695

$N = 574$; $R^2 = 0.3034$; $\bar{R}^2 = 0.2972$.

Table 11: Marriage coefficients across specifications

Group	Married (Q2)	Married (Q3b)	Mean married (Q3b)	Married (Q3c)	Other twin (Q3c)
Men	0.1993	0.0300	0.2500	0.1550	0.1250
Women	-0.0399	-0.0221	-0.0268	-0.0355	-0.0134

In a linear model, this implies the *mean-marriage* slope in (b) should be close to the average of the two slopes in (c):

$$\widehat{\beta}_{\text{mean}}^{(b)} \approx \frac{1}{2}(\widehat{\beta}_{\text{married}}^{(c)} + \widehat{\beta}_{\text{omarrried}}^{(c)}).$$

Our estimates line up: for men, $\widehat{\beta}_{\text{married}}^{(c)} + \widehat{\beta}_{\text{omarrried}}^{(c)} = 0.155 + 0.125 = 0.280$, so $\frac{1}{2}(\cdot) \approx 0.140$, which is close in magnitude to the strong family-mean effect in (b) (≈ 0.25) and explains why adding `mean.married` in (b) pulled the own-marriage coefficient toward zero.

Men. Relative to Q2, the individual marriage premium falls when adding `mean.married` (Q3b), but when we instead add `omarrried` (Q3c), both own marriage (0.155, $p = 0.004$) and co-twin marriage (0.125, $p = 0.024$) are positive and significant. Taken together, they recover a combined effect near the family-mean channel from (b). This pattern is consistent with a shared (family/selection) component to the “marriage premium” among men rather than a large individual-specific causal effect.

Women. Coefficients on both `married` and `omarrried` remain small and statistically insignificant, echoing Q2–Q3(b). Controlling for the family mean or the co-twin’s status does not reveal a meaningful association between marriage and wages for women in this sample.

Bottom line. Replacing `mean.married` with `omarrried` leaves the substantive story intact: the apparent premium in men is largely accounted for by shared (family-level) factors that influence both twins; for women, marriage has no robust association with wages across specifications.

Q4. Three Ways to Get the Within Estimator (men only)

(a) Within estimator via family demeaning

Setup. Restrict the sample to *male* twins ($N = 500$). For each family f with twins $t \in \{1, 2\}$, apply the within (demeaning) transformation

$$\tilde{y}_{ft} \equiv y_{ft} - \bar{y}_f, \quad \tilde{x}_{ft} \equiv x_{ft} - \bar{x}_f,$$

where $x_{ft} \in \{\text{educ}, \text{exp}^2, \text{married}\}$ and $y_{ft} = \text{lw}_{ft}$. Because experience is defined as $\text{exp} = \text{age} - \text{educ} - 6$ and identical twins are the same age, $\widetilde{\text{exp}}_{ft} = -\widetilde{\text{educ}}_{ft}$ within family. Hence $\widetilde{\text{exp}}$ is perfectly collinear with $\widetilde{\text{educ}}$ (corr. = -1), so we drop $\widetilde{\text{exp}}$ and keep $\widetilde{\text{exp}}^2$.

Model. We estimate the within regression (no intercept after demeaning):

$$\tilde{\mathbf{w}}_{ft} = \gamma_1 \tilde{\mathbf{educ}}_{ft} + \gamma_3 \widetilde{\mathbf{exp}^2}_{ft} + \gamma_4 \tilde{\mathbf{married}}_{ft} + u_{ft}^W,$$

with HC1 robust standard errors.

Table 12: Within-family (demeaned) regression for men (HC1 robust)

Variable	Coef.	SE(HC1)	<i>t</i>	<i>p</i>
$\tilde{\mathbf{educ}}$	0.0851	0.0369	2.3087	0.0210
$\widetilde{\mathbf{exp}^2}$	0.0004	0.0010	0.4402	0.6598
$\tilde{\mathbf{married}}$	0.0247	0.0470	0.5261	0.5988
$N = 500; R^2 = 0.0357.$				

Interpretation

- The within estimator compares twins *within the same family*, removing all time-invariant shared family factors.
- A one-year within-pair difference in schooling is associated with about 0.085 higher log wages for the more educated brother (significant at 5%).
- The coefficients on the demeaned quadratic experience term and marital status are small and imprecise, suggesting little additional *within-pair* wage difference from these factors once shared family effects are removed.

(b) Fixed Effects

Model. We now estimate the wage equation for male twins using a *family fixed effects* model, which controls for all unobserved, time-invariant family characteristics shared by the twins (e.g., parental background, neighborhood). The regression is

$$\mathbf{lw}_{ft} = \alpha_f + \beta_1 \mathbf{educ}_{ft} + \beta_3 \mathbf{exp}_{ft}^2 + \beta_4 \mathbf{married}_{ft} + u_{ft},$$

where α_f denotes the family fixed effect. As in part (a), we omit $\mathbf{exp}_{ft}$ because, for identical twins of the same age, $\widetilde{\mathbf{exp}}_{ft}$ is perfectly collinear with $\widetilde{\mathbf{educ}}_{ft}$ when conditioning on α_f .

Interpretation

- The family FE specification compares twins within the same family while allowing each family to have its own intercept α_f . This removes all time-invariant shared background

Table 13: Fixed effects (men): family FE with HC1 robust SEs

Variable	Coef.	SE(HC1)	<i>t</i>	<i>p</i>
Education	0.0851	0.0523	1.6276	0.1049
Experience ²	0.0004	0.0014	0.3103	0.7566
Married	0.0247	0.0666	0.3709	0.7110
$N = 500$; $R^2_{\text{within}} = 0.0357$; $R^2_{\text{overall}} = 0.7490$.				

factors.

- The schooling coefficient ($\hat{\beta}_1 \approx 0.085$) is positive but not statistically significant at conventional levels ($p \approx 0.105$); economically, it is close to the within (demeaned) estimate from part (a), indicating both approaches recover the same within estimator.
- The coefficients on **exp**² and **married** are small and imprecise, suggesting limited additional *within-family* wage differences associated with these covariates once shared family effects are absorbed.
- The reported R^2_{within} (≈ 0.036) summarizes variation explained *within families*; the much larger R^2_{overall} (≈ 0.749) reflects that most cross-family wage differences are captured by the family intercepts α_f .

(c) Control Function (within by adding family means)

Model. We estimate the *control function* version of the within estimator for *male* twins by regressing log wages on the individual covariates and their family means:

$$\text{lw}_{ft} = \beta_1 \text{educ}_{ft} + \beta_3 \text{exp}_{ft}^2 + \beta_4 \text{married}_{ft} + \delta_1 \text{educ}_f + \delta_3 \overline{\text{exp}^2}_f + \delta_4 \overline{\text{married}}_f + u_{ft},$$

using HC1 robust standard errors. Writing $x_{ft} = (x_{ft} - \bar{x}_f) + \bar{x}_f$ shows that including $\bar{x}_f$ absorbs shared family background; the coefficients on the *individual* regressors ($\beta_1, \beta_3, \beta_4$) coincide with the slopes from the demeaned and family-fixed-effects regressions.

Equivalence check. The estimated slopes on the *individual* regressors match those from parts (a) and (b):

Table 14: Control function (men): individual covariates and family means (HC1 robust)

Variable	Coef.	SE(HC1)	t	p
Education	0.0851	0.0689	1.2352	0.2168
Experience ²	0.0004	0.0018	0.2366	0.8130
Married	0.0247	0.0987	0.2504	0.8023
Family mean education	0.0030	0.0700	0.0431	0.9656
Family mean experience ²	-0.0003	0.0018	-0.1426	0.8866
Family mean married	0.3590	0.1199	2.9934	0.0028
$N = 500; R^2 = 0.1631.$				

Table 15: Slopes on individual covariates are identical across (a), (b), and (c)

Variable	(a) Demeaned	(b) Family FE	(c) Control fn.
Education	0.0851	0.0851	0.0851
Experience ²	0.0004	0.0004	0.0004
Married	0.0247	0.0247	0.0247

Interpretation and algebraic link

Because identical twins share age, $\widetilde{\text{exp}}_{ft} = -\widetilde{\text{educ}}_{ft}$, so **exp** drops from the within/FE equations and only **exp**² varies within family. Decompose each covariate:

$$x_{ft} = (x_{ft} - \bar{x}_f) + \bar{x}_f.$$

Substituting into a linear model yields

$$\mathbf{1}w_{ft} = \underbrace{\beta^W}_{\text{within slope}} (x_{ft} - \bar{x}_f) + \underbrace{\delta}_{\text{between slope}} \bar{x}_f + \varepsilon_{ft}.$$

Therefore, the coefficient on x_{ft} in (c) equals the within slope β^W recovered in (a) (demeaning) and (b) (family FE). Numerically, our estimates confirm this equivalence (Tables 12, 13, 14, 15). Economically, the within-family schooling effect is about 0.085 log points per year, while within-family variation in **exp**² and **married** explains little. The positive, significant family-mean coefficient on $\overline{\text{married}}_f$ captures a *shared* marriage component (selection/common factors) rather than an individual-specific premium.